NTIRE 2026 3D Scene Relighting Challenge Factsheet

Team 3dv lowlight

Haojie Guo

1 Team Information

Team name: 3dv lowlight
Member(s): Haojie Guo

2 Method Overview

Our final submission is a three-stage low-light 3D Gaussian Splatting (3DGS) pipeline designed around one central principle: in low-light scenes, geometry should be stabilized before aggressive relighting is introduced. Accordingly, the first two stages focus on geometry construction and refinement, while the third stage freezes geometry and optimizes appearance only.

The retained pipeline is:

1. **Geometry bootstrap** : fixed-pose COLMAP sparse initialization with weak depth and multiview regularization.
2. **Sparse-anchored geometry refinement** : no-densify warm-start refinement with weak structure prior, weak depth prior, and selective sparse-guided geometric anchoring.
3. **Decoupled appearance relighting** : geometry-frozen relighting with calibrated proxy supervision, confidence-aware reconstruction weighting, luminance-only illumination modulation, and additive chroma correction in YCbCr space.

This decomposition was chosen because directly optimizing geometry and aggressive low-light restoration together was less stable in our experiments. In contrast, the retained design keeps the geometry stage conservative and postpones most brightness/color restoration to a final appearance-only stage.

3 Main Technical Ideas

We summarize the retained method with three main ideas.

1. **Fixed-pose sparse bootstrap instead of monocular-depth-only initialization.** Rather than relying on a strong monocular-depth initialization, we explicitly write the official training poses into a manual COLMAP model, run feature extraction and matching on deterministic supervision images, and triangulate sparse points under known poses. These sparse points are then used as a reliable geometric bootstrap.
2. **Quality-aware soft sparse anchoring during geometry refinement.** In Stage 2, COLMAP sparse points are not only used for initialization; they are reused as weak training-time anchors. This anchor is selective rather than hard: support weights depend on sparse-point quality and local density, and the loss uses a robust Charbonnier-style distance instead of a hard nearest-point constraint.
3. **Decoupled luminance–chroma relighting.** In Stage 3, relighting is factorized into luminance recovery and chroma correction. We first apply a learned scalar illumination field only to the Y channel, and then predict an additive chroma residual on the Cb/Cr channels. This design reduces the tendency of the model to entangle brightness restoration with color drift.

4 Pipeline and Main Modules

4.1 Stage 1: Geometry bootstrap

This stage builds the initial 3DGS geometry.

- We export deterministic supervision images from the training views using the same low-light augmentation pipeline in evaluation mode.
- We write the official training poses into a manual COLMAP model and run feature extraction, exhaustive matching, triangulation, and point filtering under fixed poses.
- The resulting sparse points are stored in `auxiliaries/colmap_sparse/points.npy` and used to initialize the Gaussian means together with a small random point set.
- Stage 1 uses standard RGB reconstruction as the main supervision, with weak exposure control, a weak Marigold depth prior, and a weak multiview reprojection term.
- Densification is enabled only in the early period to grow missing support, and then stopped before the later refinement stages.

In practice, this stage provides a more stable geometric bootstrap than directly trusting a strong monocular-depth-based initialization in very dark regions.

4.2 Stage 2: Sparse-anchored geometry refinement

Stage 2 warm-starts from Stage 1 and disables densification. Its purpose is to refine geometry while preventing the representation from drifting under low-light ambiguity.

In addition to RGB reconstruction, this stage uses a weak depth prior, a structure prior, and a sparse-guided regularization term. Let μ_i denote a sampled active Gaussian center and let $\{p_{ij}\}_{j=1}^{k}$ be its k nearest sparse support points. The Stage-2 sparse loss can be summarized as

$$\mathcal{L}_{\text{sparse}} = \frac{1}{M}\sum_{i=1}^{M}\rho\left(\left\|\mu_i - \sum_{j=1}^{k} w_{ij}p_{ij}\right\|_2\right), \tag{1}$$

where $\rho(\cdot)$ is a Charbonnier-style robust penalty and the normalized weights w_{ij} depend on both distance and sparse-point reliability.

More specifically, sparse support is made selective by combining:

- **track length**, favoring points observed by more views,
- **reprojection error**, down-weighting less reliable points,
- **local sparse density**, compensating for highly uneven sparse support.

As a result, Stage 2 acts as a *soft geometric anchor*: it nudges active Gaussians toward reliable sparse support without over-constraining the representation.

4.3 Stage 3: Decoupled appearance relighting

Stage 3 warm-starts from Stage 2 and freezes geometry (Gaussian means, rotations, and scales). Only appearance-related parameters continue to update.

The relighting head is explicitly decoupled in YCbCr space. Given a rendered RGB image I, we first convert it to YCbCr and apply a learned luminance gain only to the Y channel:

$$I_{\text{base}} = \text{YCbCr}^{-1}(g_Y \cdot Y,\ C_b,\ C_r), \qquad g_Y = 2\sigma(a_Y). \tag{2}$$

Then an additive chroma residual is predicted for the chroma channels:

$$I_{\text{out}} = \text{YCbCr}^{-1}(Y',\ C_b + \Delta C_b,\ C_r + \Delta C_r). \tag{3}$$

This stage is supervised by an adaptively calibrated proxy target. The proxy target is built from a global brightness scaling branch plus a shadow-aware blending branch, and its target brightness is calibrated according to clipped grayscale statistics and highlight statistics of the current low-light view. Reconstruction losses are further reweighted by confidence-aware maps that emphasize dark informative regions while suppressing uncertain or already bright regions. In our implementation, these confidence weights are derived from shadow emphasis, rendered alpha confidence, structure confidence, and bright-region suppression.

5 Training and Inference Settings

5.1 Training schedule

The retained schedule is:

- **Stage 1:** 15,000 iterations
- **Stage 2:** 5,000 iterations
- **Stage 3:** 5,000 iterations

Shared retained settings include:

- initial Gaussian count: 100,000
- SH degree: 3
- scene scale: 2.0
- Stage 1 uses early densification only; Stage 2 and Stage 3 disable densification
- Stage 2 uses weak sparse-guided regularization with robust k-NN support
- Stage 3 freezes geometry and optimizes luminance/chroma appearance heads only

5.2 Inference

Inference is performed by loading a trained checkpoint and rendering the test views with `eval.py`. The final submission images are the rendered `test/*.png` outputs from the retained third stage.

The evaluation script also supports optional metric computation (PSNR / SSIM / LPIPS when ground-truth images are available).

6 External Data, Pretrained Models, and Additional Priors

No additional paired relighting dataset is used for training. The optimization of the retained pipeline is performed on the provided challenge scenes only.

The method uses the following pretrained or precomputed priors:

- **Marigold depth prior:** a pretrained monocular depth estimator used to generate per-view depth priors stored in `auxiliaries/depth`.
- **Structure prior:** a CIConv-based illumination-invariant structure map stored in `auxiliaries/structure`.
- **Sparse prior:** fixed-pose COLMAP sparse points and metadata stored in `auxiliaries/colmap_sparse`.

These priors are used conservatively. In particular, the depth prior is treated as a weak regularizer rather than a hard initialization target, while the sparse COLMAP prior plays the dominant role in geometric stabilization.

7 Runtime and Hardware Environment

- OS: Ubuntu 22.04
- Python: 3.10
- PyTorch: 2.5.1 + CUDA 12.1
- Main package dependencies include: `torch`, `torchvision`, `gsplat`, `omegaconf`, `PyYAML`, `tqdm`, `Pillow`, `jaxtyping`, and `rich`
- Training was conducted on a single AutoDL cloud GPU instance (vGPU-32GB class)

Approximate runtime for one scene:

- Marigold depth extraction: about 30 seconds
- structure-prior extraction: less than 10 seconds
- fixed-pose COLMAP sparse triangulation: about 3 minutes
- three training stages in total (15k + 5k + 5k iterations): about 35–40 minutes

8 Code Link and Entry Points

GitHub repository: https://github.com/Wayneee42/3dv-lowlight

Main code entry points are:

- `train.py`: unified training entry point for all retained stages
- `eval.py`: checkpoint rendering and evaluation entry point
- `config/stage4_tuned_colmap/`: Stage-1 configuration files
- `config/stage5b_ft/`: Stage-2 configuration files
- `config/stage6_adaptive_chroma_ycbcr_additive/`: Stage-3 configuration files
- `tools/build_fixed_pose_colmap_sparse_init.py`: fixed-pose COLMAP sparse prior generation
- `tools/extract_marigold_depth.py`: depth prior extraction
- `tools/extract_structure_prior.py`: structure prior extraction

The retained full pipeline is:

1. extract Marigold depth priors
2. extract structure priors
3. generate fixed-pose COLMAP sparse points
4. train `stage4_tuned_colmap`
5. train `stage5b_ft`
6. train `stage6_adaptive_chroma_ycbcr_additive`

9 Summary

Our final system is a staged low-light 3DGS pipeline that first stabilizes geometry and then performs appearance-only relighting. Its main characteristics are:

- fixed-pose COLMAP sparse initialization under the official poses,
- quality-aware weak sparse anchoring during geometry refinement,
- a geometry-frozen final appearance stage with calibrated proxy supervision,
- luminance-only illumination gain and additive YCbCr chroma residual for decoupled relighting.

Overall, the retained design emphasizes stability, selective use of priors, and explicit separation between geometry refinement and low-light appearance restoration.